

WASP-43b

R-1825

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_220307 · created 2026-08-03T05:09:48+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459645.859 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.02
Transit depth (Rp/Rs)^2	2.37 +/- 0.62 [%]
Orbital Inclination (inc)	81.32 +/- 1.71
Airmass coefficient 1 (a1)	1.121 +/- 0.013
Airmass coefficient 2 (a2)	-0.0691 +/- 0.0067
Scatter in the residuals of the lightcurve fit is	1.13 %
Best Comparison Star	None
Optimal Aperture	3.99
Optimal Annulus	7.74
Transit Duration (day)	0.047 +/- 0.0071

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.154 $\pm$ 0.02 vs 0.1594 (deviation 0.27 σ)
Duration fit vs published	0.047 d vs 0.0512 d (deviation 0.6 σ)
Mid-time O–C	-2.8 min
Depth SNR	7.7
Residual scatter	1.13 %

Light curve

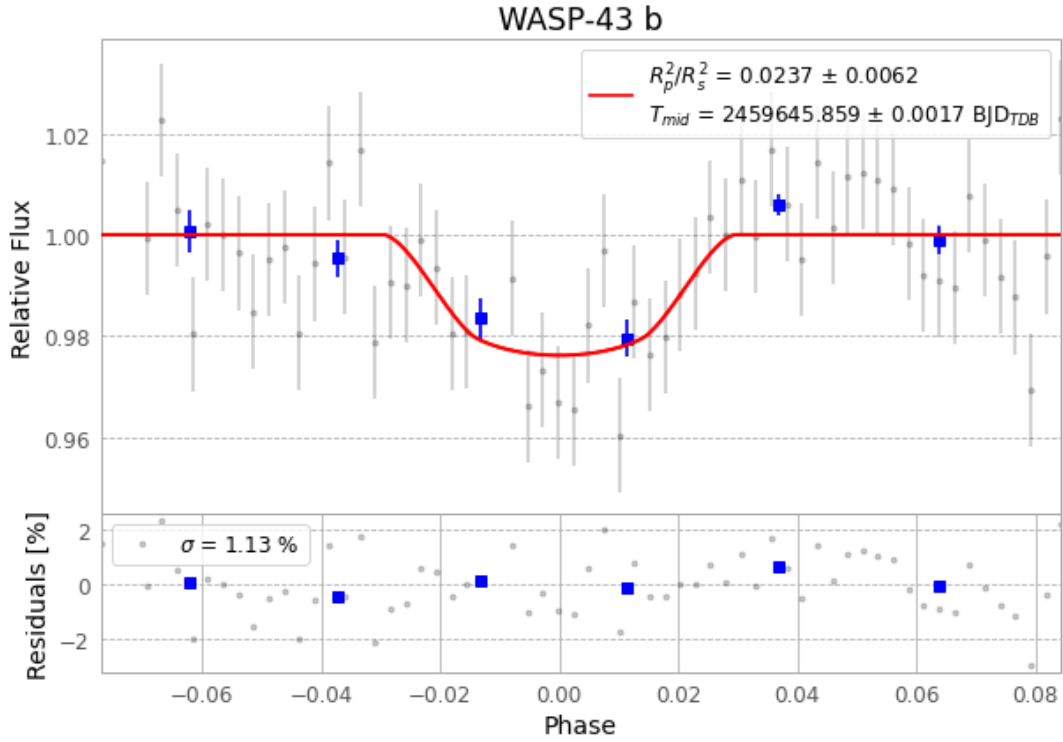


Figure 1 — FinalLightCurve_WASP-43 b_2022-03-06.png (SHA-256 a94a42e55cf6525e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [319, 255], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 acbcc1f7986b875c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

62 FITS frames (41 MB), WASP-43220307065712.FITS → WASP-43220307100613.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-220307.log (8e9a5e48dfbd...), FinalLightCurve_WASP-43 b_2022-03-06.png (a94a42e55cf6...), FinalLightCurve_WASP-43 b_2022-03-06.csv (b57033079e08...)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 05:09Z.